CMIP
Coupled Model Intercomparison Project

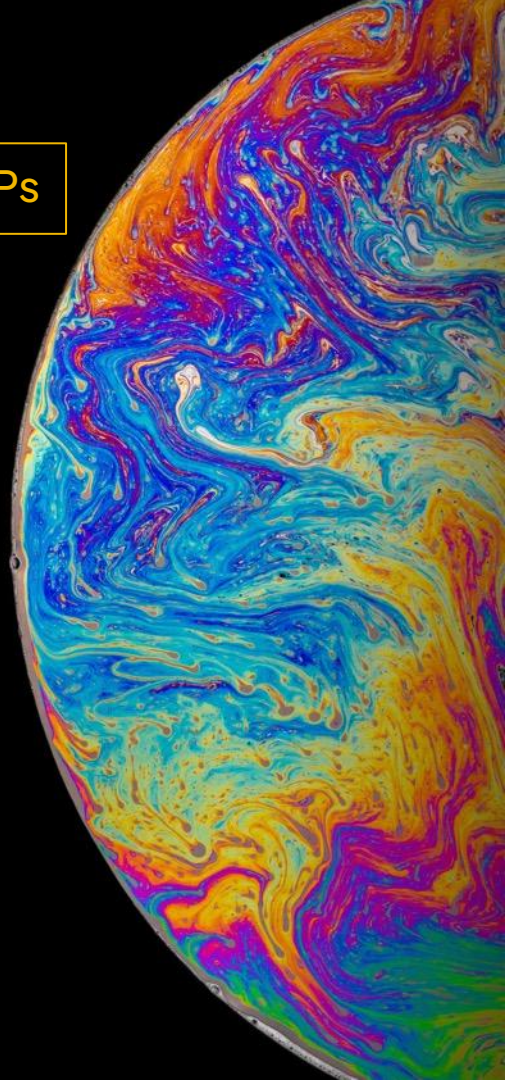
The Deck

Community MIPs

AFT

REF

How to start?



CMIP7 and Rapid Evaluation Framework (REF)

11 November 2026 - Online Seminar - Fresh Eyes on CMIP

CMIP7 Research Questions

CMIP Coupled Model Intercomparison Project

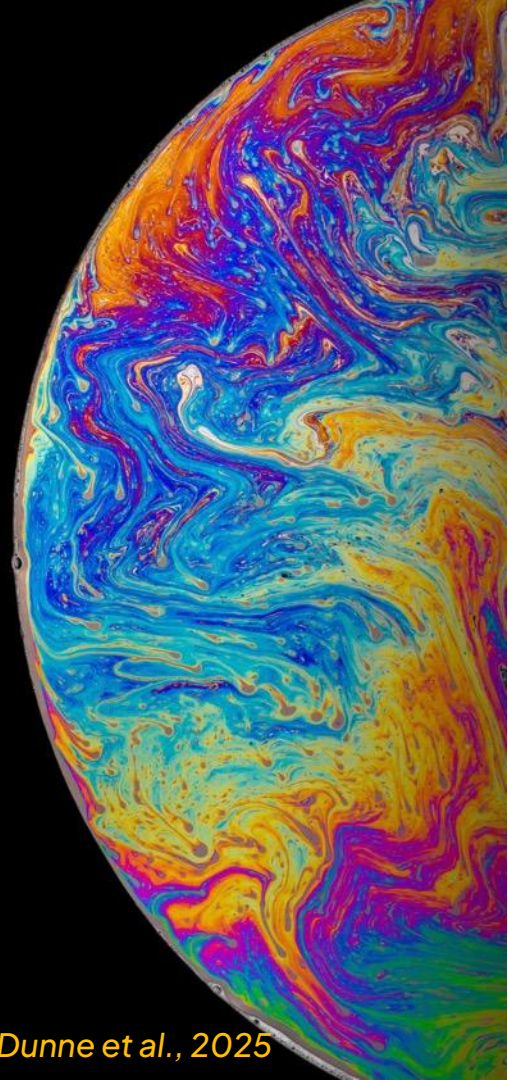








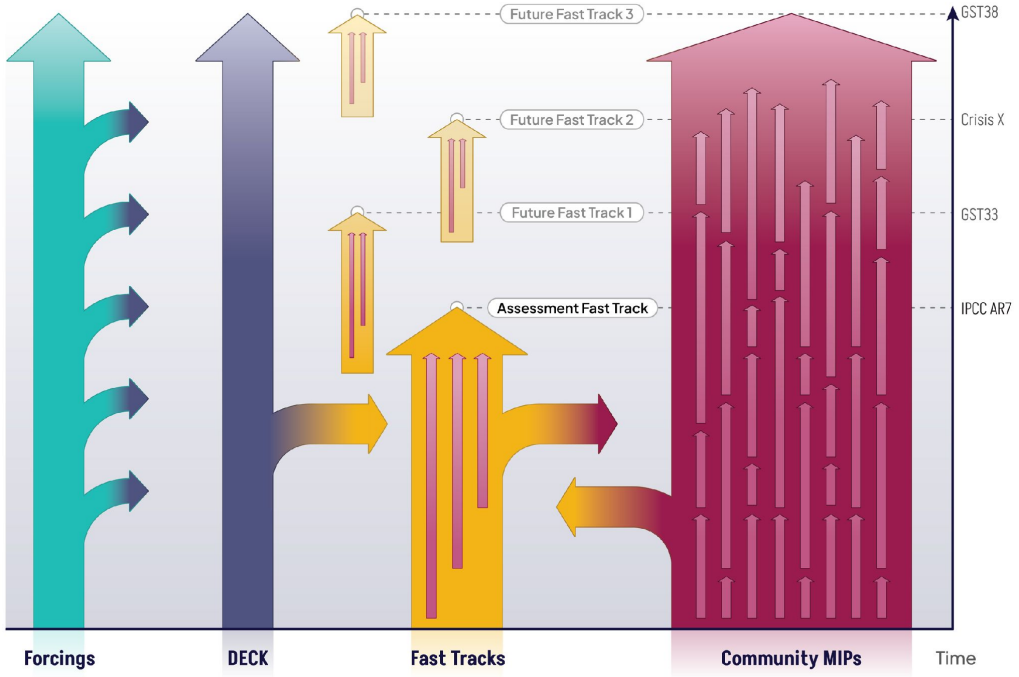
Question
Patterns of sea surface change: How will tropical ocean temperature patterns co-evolve with those at higher latitudes?
Changing weather: How will dangerous weather patterns evolve?
Water-carbon-climate nexus: How will Earth respond to human efforts to manage the carbon cycle?
Tipping Points: What are the risks of triggering irreversible changes across possible climate trajectories?



Dunne et al., 2025

CMIP7 Experiment Design

Evolved CMIP7 design into a continued **DECK**, regular updates of **forcings**, targeted experiment sets for the "**Assessment Fast Track**", and ongoing science activities through **community MIPs**.



Dunne et al., 2025
or
https://wcrp-cmip.org/cmip-phases/cmip7/

CMIP

The DECK

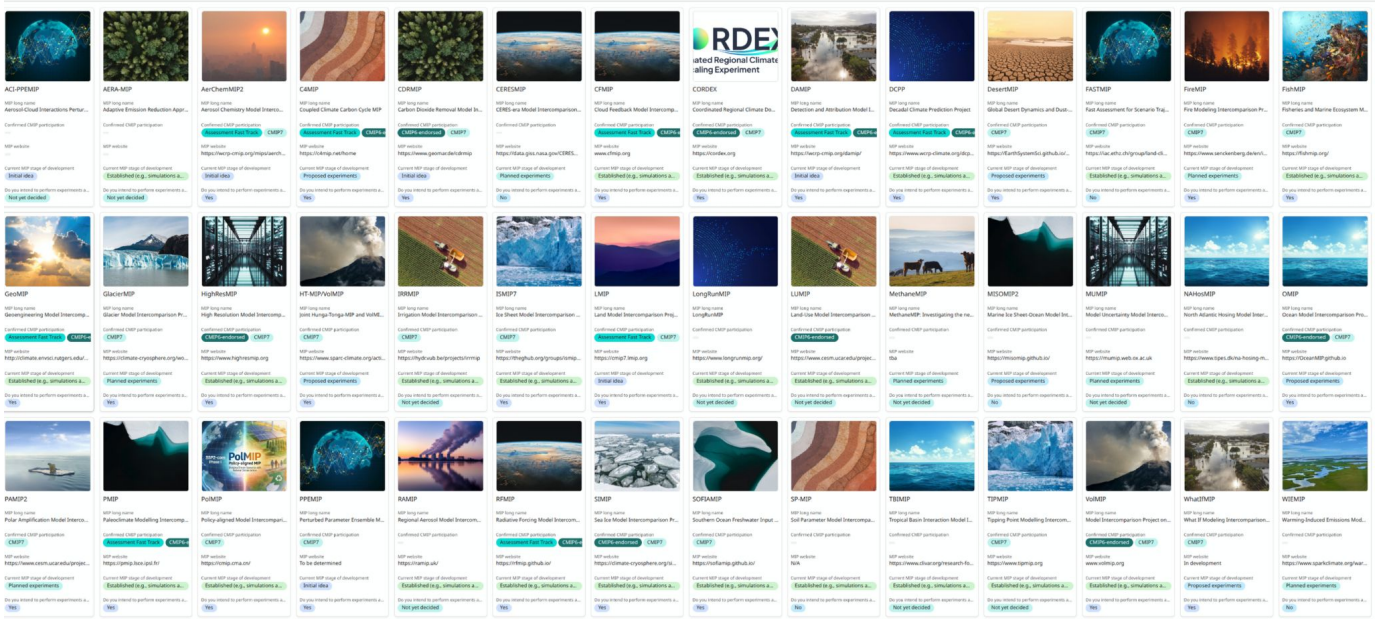
- Running the Diagnostic, Evaluation and Characterization of Klima (DECK) will remain the “entry card” for CMIP including the Assessment Fast Track.

- The following experiments are included to DECK*:
  - amip
  - abrupt 4×CO_2
  - piControl (and esm-piControl for ESMs)
  - 1pctCO_2
  - **historical**
  - **esm-hist (for ESMs only)**
  - **piClim-control**
  - **piClim-anthro**
  - **piClim-4×CO_2**

**This expanded mandatory DECK is intended to allow for more complete description and characterization.*

Community-driven science

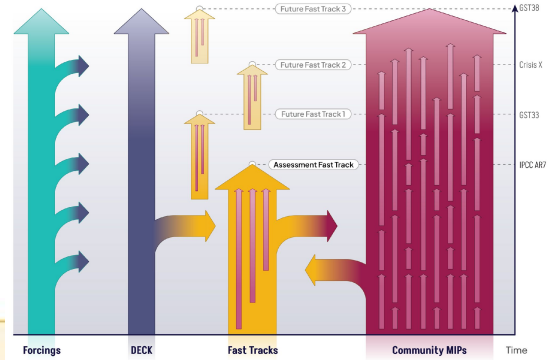
- CMIP Panel will no longer endorse MIPs, but provide best practice guidelines
- Currently 42 registered MIPs (*https://wcrp-cmip.org/mips/#registered-mips*)
- MIPs are encouraged to not re-run CMIP6 experiments with new models, to avoid duplication and consider their carbon footprint



CMIP

Assessment Fast Track (AFT)

AFT is a subset of CMIP7 experiments drawn from the community MIPs intended to support upcoming policy assessments and downstream climate services applications.



Prediction & Projection

DCPP
- dcppB-forecast-cmip6

ScenarioMIP
- (esm-)scen7-h
- (esm-)scen7-hl
- (esm-)scen7-m
- (esm-)scen7-ml
- (esm-)scen7-l
- (esm-)scen7-ln
- (esm-)scen7-vl
- Highest priority scenario extensions, as defined in van Vuuren et al., 2025

Process Understanding

Attribution

AerChemMIP
- (esm-)scen7-h-Aer
- (esm-)scen7-h-AQ
- (esm-)scen7-vl-Aer
- (esm-)scen7-vl-AQ
- hist-piAer
- hist-piAQ
- piClim-CH4
- piClim-N2O
- piClim-NOx
- piClim-ODS
- piClim-SO2

Characterisation

GeoMIP
- g7-1p5K-sai

PMIP
- abrupt-127k

DAMIP
- hist-nat
- hist-aer
- hist-GHG

LMIP
- land-hist

C4MIP
- 1pctCO2-bgc
- 1pctCO2-rad
- esm-flat10
- esm-flat10-cdr
- esm-flat10-zec

CFMIP
- amip-p4k
- amip-piForcing
- abrupt-2xCO2
- abrupt-0p5CO2

RFMIP
- piClim-histaer
- piClim-aer
- piClim-histall

Scientific questions

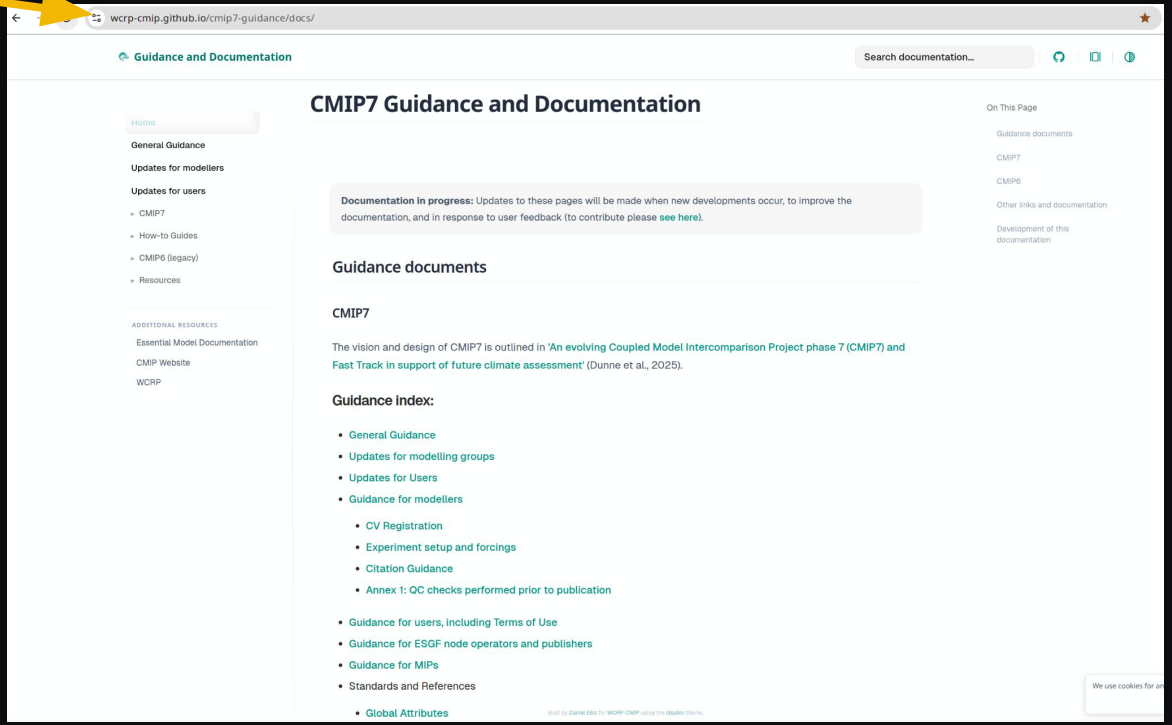
- Patterns of sea surface change
- Changing weather
- Water-carbon-climate nexus
- Tipping Points

Assessment Fast Track

CMIP

Interested in working with CMIP7 data?

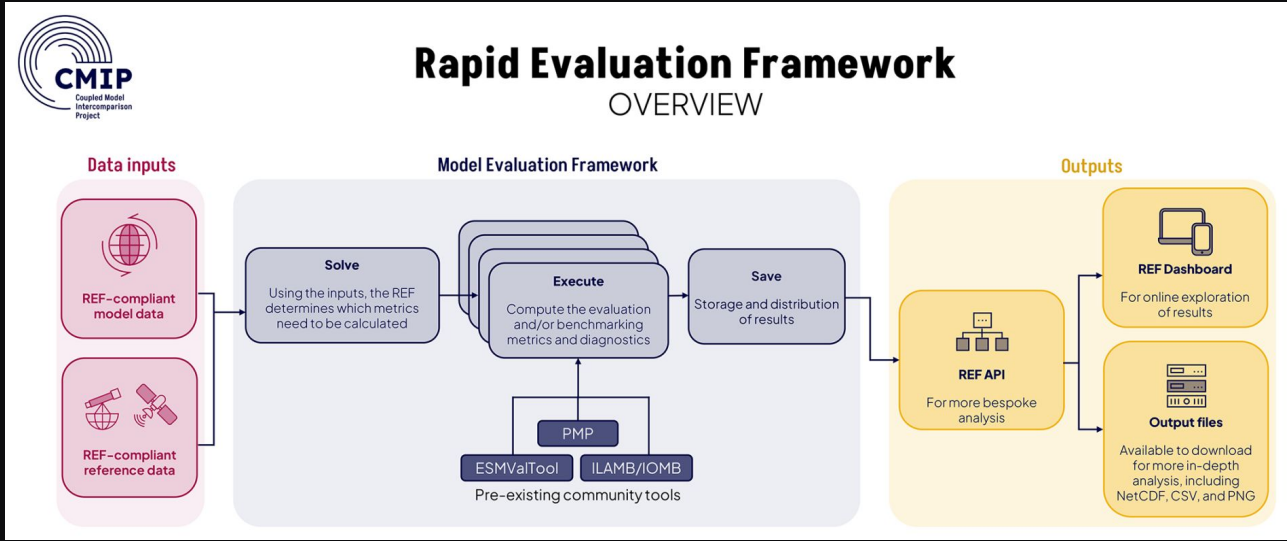
https://wcrp-cmip.github.io/cmip7-guidance/docs/



Rapid Evaluation Framework (REF)

community-driven platform for benchmarking and performance assessment of ESMs, constructed to meet the near-term needs of the CMIP7 Assessment Fast Track

Hoffman et al., 2026



https://dashboard.climate-ref.org/

Interested in CMIP7?

- Take your time and read about CMIP7 data
  - *Brief review: https://wcrp-cmip.org/cmip-phases/cmip7/*
  - *Guidance and Documentation: https://wcrp-cmip.github.io/cmip7-guidance/docs/*
  - *Scientific papers: Dunne et al., 2025 for CMIP7 design, Hoffman et al. 2026 for REF;*

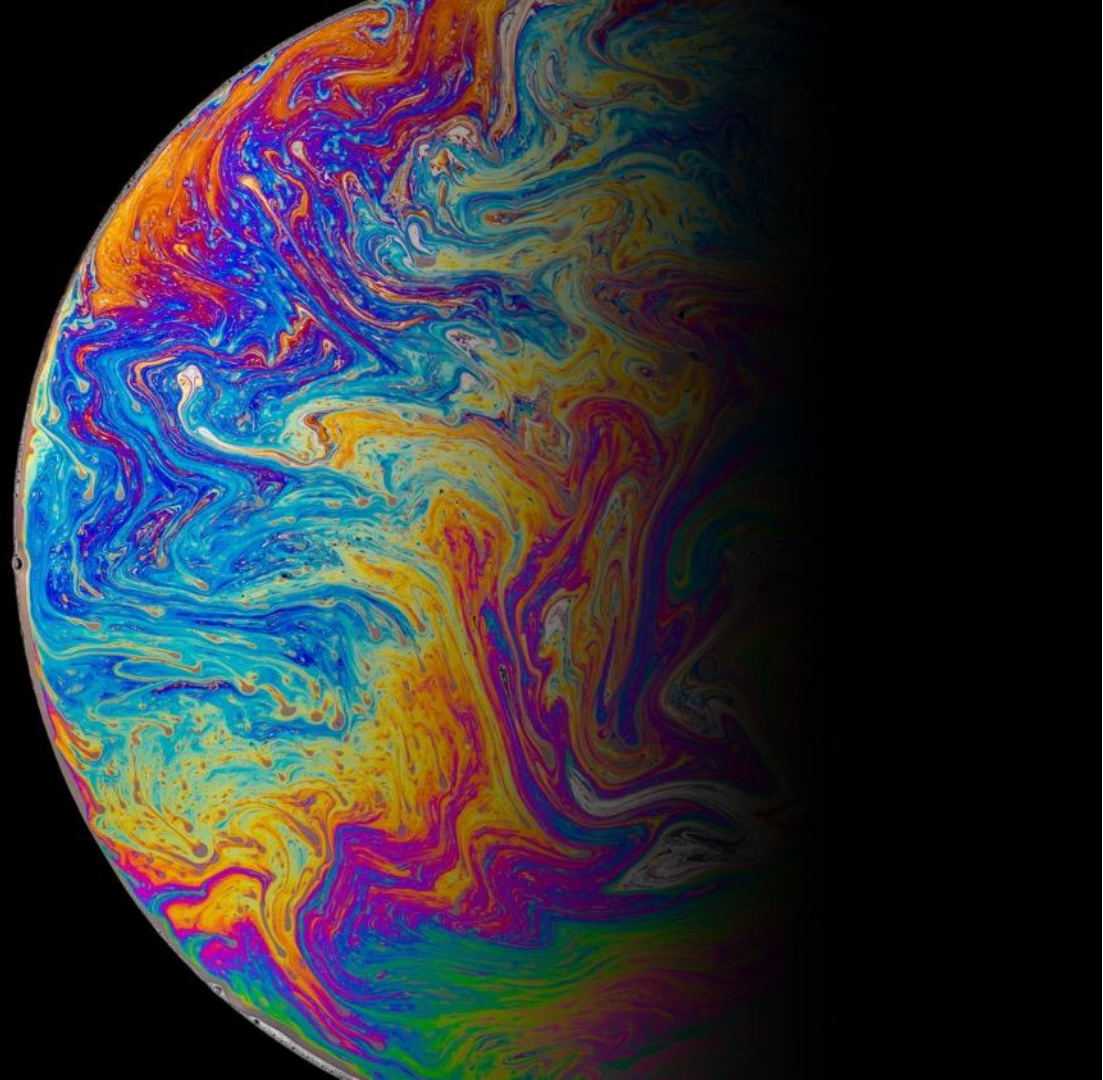
- Discover CMIP7 data on ESGF: *https://metagrid-ceda.east.esgf.io/search*

- REF webpage: *https://dashboard.climate-ref.org/*

- Keep track of upcoming papers: Geoscientific Model Development (GMD) Special Issue **CMIP7 scientific objectives, experimental design, and organization**

- Keep track of upcoming events: e.g. EGU 2027, session on “Early results from Coupled Model Intercomparison Project Seventh phase (CMIP7)”, CL4.3





CMIP

Coupled Model
Intercomparison
Project